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SUB NAME: QUERY PROCESSING FOR DATA SCIENCE

CO3 ASSESSMENT TOOL 3

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Question 1 – Customer Dataset Integration

Three customer datasets from Online, Retail, and Support systems contain inconsistent names, phone numbers, email addresses, dates, duplicate records, and missing values.

Task: Using Python/Pandas, clean and integrate the datasets by identifying invalid values, formatting data, removing duplicates, applying fuzzy matching and RegEx validation, standardizing fields, and saving the final master dataset in CSV and JSON formats. Create a reusable cleanup script and test it with a new dataset.

AIM: To clean and integrate customer datasets obtained from Online, Retail, and Support systems using Python and Pandas.

OBJECTIVE: To clean and integrate customer datasets obtained from Online, Retail, and Support systems using Python and Pandas. The process includes identifying invalid values, standardizing names, phone numbers, emails, dates and cities, removing duplicate records, performing fuzzy matching, validating data using Regular Expressions, and generating a final master customer dataset in CSV and JSON formats.

DATA ISSUES IDENTIFIED:

- Inconsistent customer name formats
- Different phone number formats
- Invalid email addresses
- Different date formats
- Duplicate customer records
- Missing values
- Different city names and spellings
- Similar customer names representing the same person.

... CUSTOMER DATASET INTEGRATION

Datasets Loaded Successfully!

Online Dataset:

	CustomerID	Name	Phone	Email	Date	City
0	101	John Smith	+91-98765-43210	JOHN.SMITH@GMAIL.COM	15/01/2026	Chennai
1	102	Priya Kumar	9876543211	priya.kumar@gmail.com	2026-02-10	Chennai
2	103	Rahul Raj	98765 43212	rahul.raj@gmail.com	12-03-2026	Bangalore
3	104	Anita Joseph	98765-43213	anita.joseph@gmail	2026/04/05	Chennai
4	105	Arun Kumar	9876543214	arun.kumar@gmail.com	05/05/2026	Madurai
5	101	John Smith	+91 9876543210	john.smith@gmail.com	2026-01-15	Chennai

Retail Dataset:

	CustomerID	Name	Phone	Email	Date	City
0	201	Jon Smith	9876543210	john.smith@gmail.com	2026-01-15	CHENNAI
1	202	Priya Kumar	+91-98765-43211	PRIYA.KUMAR@GMAIL.COM	10/02/2026	Chennai
2	203	Rahul Raj	98765-43212	rahul.raj@gmail.com	2026-03-12	Bengaluru
3	204	Meena Devi	98765 43215	meena.devi@gmail.com	2026-06-01	Madurai
4	205	Karthik Rao	9876543216	karthik.rao@gmail.com	07/06/2026	Chennai

Support Dataset:

	CustomerID	Name	Phone	Email	Date	City
0	301	John Smit	+91 98765 43210	john.smith@gmail.com	15-Jan-2026	Chennai
1	302	Priya Kumar	9876543211	priya.kumar@gmail.com	2026-02-10	CHENNAI
2	303	Rahul Raj	9876543212	rahul.raj@gmail.com	12/03/2026	Bangalore
3	304	Meena Devi	9876543215	meena.devi@gmail.com	01-06-2026	MADURAI
4	305	Karthik Rao	NaN	karthik.rao@gmail.com	2026/06/07	Chennai
5	306	NaN	9876543217	invalid-email	2026-07-01	Chennai

NAME STANDARDIZATION:

Customer names were cleaned by:

- Removing leading and trailing spaces
- Removing unnecessary spaces between words
- Converting names into proper cases.

PHONE NUMBER STANDARDIZATION:

Different phone formats were converted into a common 10-digit format.

EMAIL VALIDATION USING REGEX:

Regular Expression was used to identify valid and invalid email addresses.

DATE STANDARDIZATION:

Different date formats were converted into a common format.

```
=====
... BASIC INFORMATION
=====

Online Dataset Shape: (6, 6)
Retail Dataset Shape: (5, 6)
Support Dataset Shape: (6, 6)

Standardized Column Names:
['customerid', 'name', 'phone', 'email', 'date', 'city']

Names cleaned successfully.

Phone numbers cleaned successfully.

Email validation completed.
Dates standardized successfully!

City names standardized successfully.
```

DUPLICATE REMOVAL:

Exact duplicate records were identified and removed using Pandas `drop_duplicates()`.

The three datasets were then combined and duplicate customers were identified using customer information such as email, phone number and name.

```
--- Missing values identified.

Missing Values - Online:
customerid    0
name          0
phone         0
email         1
date          0
city          0
dtype: int64

Missing Values - Retail:
customerid    0
name          0
phone         0
email         0
date          0
city          0
dtype: int64

Missing Values - Support:
customerid    0
name          1
phone         1
email         1
date          0
city          0
dtype: int64
```

```

... =====
COMBINED DATASET
=====
Total Records: 16

```

	customerid	name	phone	email	date	city
0	101	John Smith	9876543210	john.smith@gmail.com	2026-01-15	Chennai
1	102	Priya Kumar	9876543211	priya.kumar@gmail.com	2026-02-10	Chennai
2	103	Rahul Raj	9876543212	rahul.raj@gmail.com	2026-03-12	Bengaluru
3	104	Anita Joseph	9876543213	NaN	2026-04-05	Chennai
4	105	Arun Kumar	9876543214	arun.kumar@gmail.com	2026-05-05	Madurai
5	201	Jon Smith	9876543210	john.smith@gmail.com	2026-01-15	Chennai
6	202	Priya Kumar	9876543211	priya.kumar@gmail.com	2026-02-10	Chennai
7	203	Rahul Raj	9876543212	rahul.raj@gmail.com	2026-03-12	Bengaluru
8	204	Meena Devi	9876543215	meena.devi@gmail.com	2026-06-01	Madurai
9	205	Karthik Rao	9876543216	karthik.rao@gmail.com	2026-06-07	Chennai
10	301	John Smit	9876543210	john.smith@gmail.com	2026-01-15	Chennai
11	302	Priya Kumar	9876543211	priya.kumar@gmail.com	2026-02-10	Chennai
12	303	Rahul Raj	9876543212	rahul.raj@gmail.com	2026-03-12	Bengaluru
13	304	Meena Devi	9876543215	meena.devi@gmail.com	2026-06-01	Madurai
14	305	Karthik Rao	NaN	karthik.rao@gmail.com	2026-06-07	Chennai

FUZZY MATCHING:

Fuzzy matching was performed using the **RapidFuzz** library. It identifies names that are similar but not exactly identical.

```

... =====
FUZZY MATCHING
=====

Potential Duplicate Customers:

```

	Name 1	Name 2	Similarity Score
0	john smith	jon smith	94.736842
1	john smith	john smit	94.736842
2	jon smith	john smit	88.888889

```

... =====
FINAL DATA QUALITY REPORT
=====
Original Online Records : 5
Original Retail Records : 5
Original Support Records: 6

Combined Records: 16
Final Master Records: 8

Missing Values in Final Dataset:
customerid 0
name        0
phone       0
email       0
date        0
city        0
dtype: int64

Duplicate Records in Final Dataset:
0

Final Columns:
['customerid', 'name', 'phone', 'email', 'date', 'city']

=====
CUSTOMER DATASET INTEGRATION COMPLETED
=====

```

```

> cleaned_new_data = customer_cleanup.clean_dataset(
    new_data
)

display(cleaned_new_data)

```

```

.. Starting dataset cleanup...

```

	customerid	name	phone	email	date	city
0	401	Ravi Kumar	9876543220	ravi.kumar@gmail.com	2026-08-20	Chennai
1	402	Meena Devi	9876543221	meena.devi@gmail.com	2026-08-21	Madurai
2	403	Arun Kumar	9876543222	Unknown	2026-08-22	Bengaluru
3	404	Test User	9876543223	test.user@gmail.com	2026-08-23	Chennai

CONCLUSION: The three customer datasets from Online, Retail, and Support systems were successfully cleaned and integrated using Python and Pandas. Inconsistent names, phone numbers, emails, dates and city values were standardized. Invalid emails were identified using Regular Expressions, duplicate records were removed, and fuzzy matching was used to identify similar customer names. The final master dataset was successfully generated in CSV and JSON formats. A reusable cleanup script was also created and successfully tested with a new dataset.

Question 2 – Employee Dataset Integration

An organization has employee data stored in HR, Payroll, and Attendance datasets. The datasets contain different column names, inconsistent employee names, duplicate employees, invalid email IDs, different date formats, missing department values, and salary inconsistencies.

Task: Develop a Python program to integrate the three datasets. Apply data formatting, RegEx matching, duplicate detection, fuzzy matching, normalization/standardization, outlier detection, and missing-value handling. Save the cleaned integrated dataset and test the cleanup script using new employee records.

AIM: To develop a Python-based employee dataset integration system that combines HR, Payroll, and Attendance datasets.

OBJECTIVE: To develop a Python-based employee dataset integration system that combines HR, Payroll, and Attendance datasets. The system performs data formatting, Regex-based email validation, duplicate detection, fuzzy matching, normalization, standardization, salary inconsistency checking, outlier detection, and missing-value handling. The cleaned employee dataset is saved in CSV and JSON formats and the reusable cleanup script is tested with new employee records.

DATA ISSUES IDENTIFIED:

- Different column names were used in the three datasets.
- Employee names contained extra spaces and inconsistent capitalization.
- Some employee names were slightly different, such as John Smith, Jon Smith, and John Smit.
- Some email IDs were invalid.
- Different date formats were used.
- Department values were missing for some employees.
- Duplicate employee records were present.
- Salary values needed consistency checking.
- Salary outliers needed to be detected.
- Missing values were present in employee records.

... HR Dataset:

	Employee_ID	Employee_Name	Email_ID	Department	Joining_Date
0	101	John Smith	JOHN.SMITH@COMPANY.COM	HR	15/01/2020
1	102	Priya Kumar	priya.kumar@company.com	Finance	10/02/2021
2	103	Rahul Raj	rahul.raj@company.com	IT	12-03-2022
3	104	Anita Joseph	anita.joseph@company	NaN	2023/04/05
4	105	Arun Kumar	arun.kumar@company.com	Operations	05/05/2020
5	101	John Smith	john.smith@company.com	HR	15/01/2020

Payroll Dataset:

	Emp_ID	Name	Email	Salary	Salary_Date
0	201	Jon Smith	john.smith@company.com	50000	2026-01-31
1	202	Priya Kumar	PRIYA.KUMAR@COMPANY.COM	55000	31/01/2026
2	203	Rahul Raj	rahul.raj@company.com	60000	31-01-2026
3	204	Meena Devi	meena.devi@company.com	52000	2026/01/31
4	205	Karthik Rao	karthik.rao@company.com	58000	31-Jan-2026
5	206	Arun Kumar	arun.kumar@company.com	50000	2026-01-31

*** Attendance Dataset:

	EmployeeNumber	EmployeeName	EmailAddress	Attendance_Date	Days_Present
0	301	John Smit	john.smith@company.com	01/08/2026	22
1	302	Priya Kumar	priya.kumar@company.com	2026-08-02	20
2	303	Rahul Raj	invalid-email	03-08-2026	21
3	304	Meena Devi	meena.devi@company.com	2026/08/04	19
4	305	Karthik Rao	karthik.rao@company.com	05-Aug-2026	23
5	306	Test Employee	test.employee@company.com	2026-08-06	18



COLUMN STANDARDIZATION:

The three datasets used different column names for the same information.

```
HR Columns:
['employee_id', 'employee_name', 'email_id', 'department', 'joining_date']

Payroll Columns:
['emp_id', 'name', 'email', 'salary', 'salary_date']

Attendance Columns:
['employeenumber', 'employeename', 'emailaddress', 'attendance_date', 'days_present']
```

EMPLOYEE NAME STANDARDIZATION:

Employee names were standardized by:

- Removing leading and trailing spaces.
- Removing unnecessary spaces between words.
- Converting names into proper cases.

REGEX EMAIL VALIDATION:

Regular Expression (Regex) was used to validate employee email addresses.

```

▶ email_pattern = r"^[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.[A-Za-z]{2,}$"

def clean_email(value):
    if pd.isna(value):
        return ""

    value = str(value).strip().lower()

    if re.match(email_pattern, value):
        return value

    return ""

hr["email"] = hr["email"].apply(clean_email)
payroll["email"] = payroll["email"].apply(clean_email)
attendance["email"] = attendance["email"].apply(clean_email)

print("Email validation completed!")

```

```

... Email validation completed!

```

FUZZY MATCHING:

Fuzzy matching was used to identify employee names that are similar but not exactly identical.

```

▶ hr_names = hr["employee_name"].dropna().unique()
payroll_names = payroll["employee_name"].dropna().unique()
attendance_names = attendance["employee_name"].dropna().unique()

print("Potential Similar Employee Names:")

for name1 in hr_names:
    for name2 in payroll_names:
        score = fuzz.ratio(
            name1.lower(),
            name2.lower()
        )

        if score >= 80 and name1 != name2:
            print(
                name1,
                "<-->",
                name2,
                "Similarity:",
                round(score, 2)
            )

```

```

... Potential Similar Employee Names:
    John Smith <--> Jon Smith Similarity: 94.74

```

NORMALIZATION AND STANDARDIZATION:

Employee names were normalized by converting them to lowercase, removing unnecessary characters, and standardizing spaces. A normalized employee name was used as an additional key for identifying duplicate employees.

DUPLICATE DETECTION AND REMOVAL:

Duplicate employee records were identified using Pandas `drop_duplicates()`. Exact duplicate records were removed before integrating the datasets. Additional duplicate employees were identified using email, employee name, and fuzzy matching techniques. This helped reduce duplicate employee records in the final master dataset.

```
▶ master = combined.drop_duplicates(  
    subset=["master_key"],  
    keep="first"  
)  
.copy()  
  
print("Duplicate employees removed!")  
... Duplicate employees removed!
```

SALARY INCONSISTENCY DETECTION:

Salary values were analyzed to identify inconsistencies between employee records. The minimum salary, maximum salary, and salary difference were calculated for each employee. The following values were compared:

- Minimum salary
- Maximum salary
- Salary difference

This helps identify employees whose salary values are inconsistent across records.

```

> salary_data = payroll[salary]

Q1 = salary_data.quantile(0.25)

Q3 = salary_data.quantile(0.75)

IQR = Q3 - Q1

lower_limit = Q1 - 1.5 * IQR

upper_limit = Q3 + 1.5 * IQR

print("Q1:", Q1)
print("Q3:", Q3)
print("IQR:", IQR)

print("Lower Limit:", lower_limit)
print("Upper Limit:", upper_limit)

```

```

... Q1: 50500.0
    Q3: 57250.0
    IQR: 6750.0
    Lower Limit: 40375.0
    Upper Limit: 67375.0

```

DATASET INTEGRATION:

The cleaned HR, Payroll, and Attendance datasets were combined using Pandas. The datasets were first standardized and then concatenated to create a combined employee dataset. Duplicate employee records were removed using standardized employee information. The final result was a cleaned and integrated employee master dataset.

Combined Employee Dataset:

	employee_id	employee_name	email	department	joining_date	salary	salary_date	attendance_date	days_present
0	101	John Smith	john.smith@company.com	HR	NaT	NaN	NaT	NaT	NaN
1	102	Priya Kumar	priya.kumar@company.com	Finance	NaT	NaN	NaT	NaT	NaN
2	103	Rahul Raj	rahul.raj@company.com	IT	NaT	NaN	NaT	NaT	NaN
3	104	Anita Joseph		Unknown	NaT	NaN	NaT	NaT	NaN
4	105	Arun Kumar	arun.kumar@company.com	Operations	NaT	NaN	NaT	NaT	NaN
5	201	Jon Smith	john.smith@company.com	NaN	NaT	50000.0	NaT	NaT	NaN
6	202	Priya Kumar	priya.kumar@company.com	NaN	NaT	55000.0	NaT	NaT	NaN
7	203	Rahul Raj	rahul.raj@company.com	NaN	NaT	60000.0	NaT	NaT	NaN
8	204	Meena Devi	meena.devi@company.com	NaN	NaT	52000.0	NaT	NaT	NaN
9	205	Karthik Rao	karthik.rao@company.com	NaN	NaT	58000.0	NaT	NaT	NaN
10	206	Arun Kumar	arun.kumar@company.com	NaN	NaT	50000.0	NaT	NaT	NaN
11	301	John Smit	john.smith@company.com	NaN	NaT	NaN	NaT	NaT	22.0
12	302	Priya Kumar	priya.kumar@company.com	NaN	NaT	NaN	NaT	NaT	20.0
13	303	Rahul Raj		NaN	NaT	NaN	NaT	NaT	21.0
14	304	Meena Devi	meena.devi@company.com	NaN	NaT	NaN	NaT	NaT	19.0
15	305	Karthik Rao	karthik.rao@company.com	NaN	NaT	NaN	NaT	NaT	23.0
16	306	Test Employee	test.employee@company.com	NaN	NaT	NaN	NaT	NaT	18.0

FINAL MASTER EMPLOYEE DATASET :

... Final Master Employee Dataset:

	employee_id	employee_name	email	department	joining_date	salary	salary_date	attendance_date	days_present
0	101	John Smith	john.smith@company.com	HR	Unknown	0.0	Unknown	Unknown	0.0
1	102	Priya Kumar	priya.kumar@company.com	Finance	Unknown	0.0	Unknown	Unknown	0.0
2	103	Rahul Raj	rahul.raj@company.com	IT	Unknown	0.0	Unknown	Unknown	0.0
3	104	Anita Joseph		Unknown	Unknown	0.0	Unknown	Unknown	0.0
4	105	Arun Kumar	arun.kumar@company.com	Operations	Unknown	0.0	Unknown	Unknown	0.0
5	204	Meena Devi	meena.devi@company.com	Unknown	Unknown	52000.0	Unknown	Unknown	0.0
6	205	Karthik Rao	karthik.rao@company.com	Unknown	Unknown	58000.0	Unknown	Unknown	0.0
7	303	Rahul Raj		Unknown	Unknown	0.0	Unknown	Unknown	21.0
8	306	Test Employee	test.employee@company.com	Unknown	Unknown	0.0	Unknown	Unknown	18.0

NEW EMPLOYEE DATASET:

... NEW EMPLOYEE DATASET

	Employee_ID	Employee_Name	Email	Department	Joining_Date	Salary
0	401	Ravi Kumar	RAVI.KUMAR@COMPANY.COM	IT	20/08/2026	62000
1	402	MEENA DEVI	meena.devi@gmail	NaN	2026-08-21	58000
2	403	Karthik Rao	karthik.rao@company.com	Finance	22-08-2026	59000
3	404	Priya Kumar	PRIYA.KUMAR@COMPANY.COM	HR	2026/08/23	55000
4	405	New Employee	new.employee@company.com	Operations	24-Aug-2026	61000

Next steps: [Generate code with new_employees](#) [New interactive sheet](#)

CLEANED NEW EMPLOYEE DATASET:

... Starting employee dataset cleanup...

CLEANED NEW EMPLOYEE DATASET

	employee_id	employee_name	email	department	joining_date	salary
0	401	Ravi Kumar	ravi.kumar@company.com	IT	2026-08-20	62000
1	402	Meena Devi		Unknown	2026-08-21	58000
2	403	Karthik Rao	karthik.rao@company.com	Finance	2026-08-22	59000
3	404	Priya Kumar	priya.kumar@company.com	HR	2026-08-23	55000
4	405	New Employee	new.employee@company.com	Operations	2026-08-24	61000

CONCLUSION: The HR, Payroll, and Attendance employee datasets were successfully cleaned and integrated using Python and Pandas. Inconsistent column names, employee names, email IDs, dates, and department values were standardized. Regex was used for email validation, duplicate records were identified and removed, and fuzzy matching was used to detect similar employee names. Salary inconsistencies were analyzed and the IQR method was used for salary outlier detection. Missing values were handled appropriately, and the final integrated employee dataset was generated. A reusable cleanup script was also created and successfully tested with new employee records.